Graphical method

1. $2 \times n$ game = 2 Row and n column.
2. $m \times 2$ game = m row and 2 column.

Graphical method for $m \times 2$ game:

1. Reduce the size of the pay-off matrix of player A by applying the dominance property, if it exists.
2. Let y/p_1 be the probability of selection of alternative I by player B and $1-y/1-p_1$ be the probability of selection of alternative 2 by player B.

   Derive the expected gain function of player B with respect to each of the alternatives of player A.
3. Find the value of the ~~game~~ gain, when $y=0$ and $y=1$.
4. Plot the gain function on a graph by assuming a suitable scale. Keep y on x-axis and the gain on y-axis.
5. Find the lowest intersection point in the upper boundary of the graph. i.e. Minimax point.
6. If the no. of lines passing through the minimax point is only two, from a 2×2 pay-off matrix. Go to step 8 and otherwise go to step 7.
7. Identify any two lines with opposite slopes passing through that point. Then from a 2×2 pay-off matrix.
8. Solve the 2×2 game using oddments and find the strategies for player A and B and also find the value of the game.

CamScanner

Numerical problem of m×2 game:

① consider the pay-off matrix of player A and solve optimally using graphical method.

		Player B	
		1	2
Player A	1	6	-7
	2	1	3
	3	3	1
	4	5	-1

Solution: Given

		Player B 1	Player B 2	Min	
Player A	1	6	-7	-7	
	2	1	3	1	Maximin
	3	3	1	1	Maximin
	4	5	-1	-1	
max.		6	3		
			Minimax		

$\therefore$ Maximin $\neq$ minimax

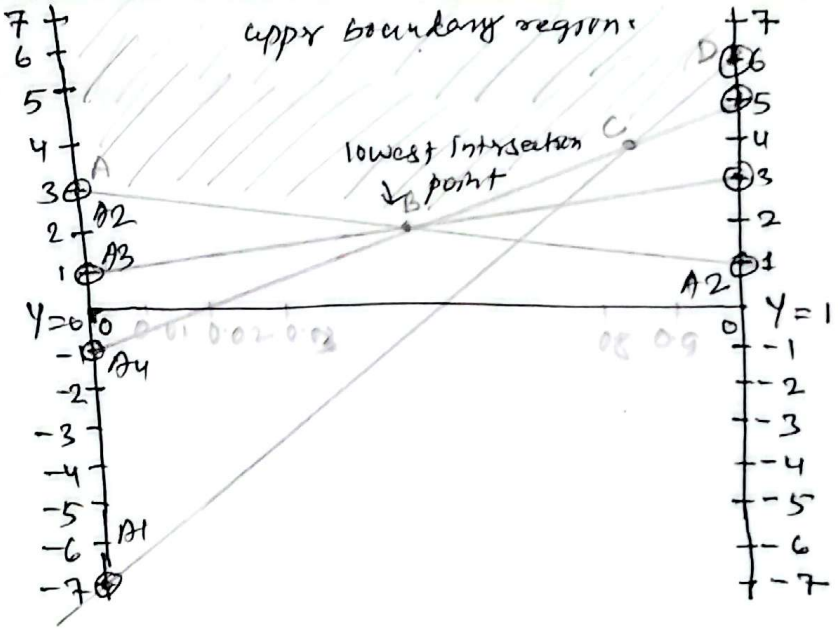
$1 \neq 3$

It is called mixed strategy

Now by using Graphical method,

Expected pay-off function and expected gain of player B

A's alternatives	B's expected pay-off function	B's expected gain	
		$y=0$	$y=1$
1	$6y-7(1-y) = 6y-7+7y = 13y-7$	-7	6
2	$y+3(1-y) = y+3-3y = -2y+3$	3	1
3	$3y+1(1-y) = 3y+1-y = 2y+1$	1	3
4	$5y-1(1-y) = 5y-1+y = 6y-1$	-1	5



upper boundary points A, B, C and D. but point B is called minimax point.

Choose any two opposite slopes, A_2 and A_4 are opposite slopes passing through minimax point.

	Player B 1	2	oddments	Probability
2	1	3	6	$\frac{6}{8} = \frac{3}{4}$
4	5	-1	2	$\frac{2}{8} = \frac{1}{4}$
oddments	4	4		

Probability $\frac{4}{4+8} = \frac{1}{2}$ $\frac{4}{8} = \frac{1}{2}$

Value of game (V) $= \frac{1\times 6 + 5\times 2}{6+2} = \frac{6+10}{8} = 2$

$\therefore$ optimum strategy of player A $= (0, \frac{3}{4}, 0, \frac{1}{4})$

" " " B $= (\frac{1}{2}, \frac{1}{2})$

value of the game (V) = 2 Ans.

Graphical method for 2×n game:

Step(1) Reduce the size of the pay-off matrix of player A by applying dominance property, if it exists.

Step(2): Let x be the probability of selection of alternative 1 by Player A and $1-x$ be the probability of selection of alternative 2 by player A.

Derive the expected gain function of Player A with respect to each of the alternatives of Player B.

Step 3: Find the value of the gain, when $x=0$ and $x=1$.

Step 4: Plot the gain function on a graph by assuming a suitable scale. Keep x on x-axis and the gain on y-axis.

Step 5: Find the highest intersection point in the lower boundary of the graph. i.e. maximin point.

Step 6: If the no. of lines passing through the maximin point is only two, from a 2×2 pay-off matrix. Go to step 8, otherwise go to step 7.

Step 7: Identify any two lines with opposite slopes passing through that point. Then from a 2×2 pay-off matrix.

Step 8: Solve the 2×2 game using oddments and find the strategies for player A and B and also value of the game.

Numerical problem

Question: Consider the pay-off matrix of Player A as shown below and solved it optimally using graphical method.

		Player B				
		1	2	3	4	5
Player A	1	-4	2	5	-6	-6
	2	3	-9	7	4	8

Solution: Given,

		Player B					R min
		1	2	3	4	5	
Player A	1	-4	2	5	-6	6	$\boxed{-6}$ Maximin
	2	3	-9	7	4	8	-9
C-max.		3	$\boxed{2}$	7	4	8	

Minimax.

since, maximin ≠ minimax, it has no saddle point, then it is mixed strategy. Now checked Dominance property.

~~Now~~

	Player B				
	1	2	3	4	5
Player A 1	-4	2	5	-6	6
2	3	-9	7	4	8
C-sum	-1	-7	12	-2	14

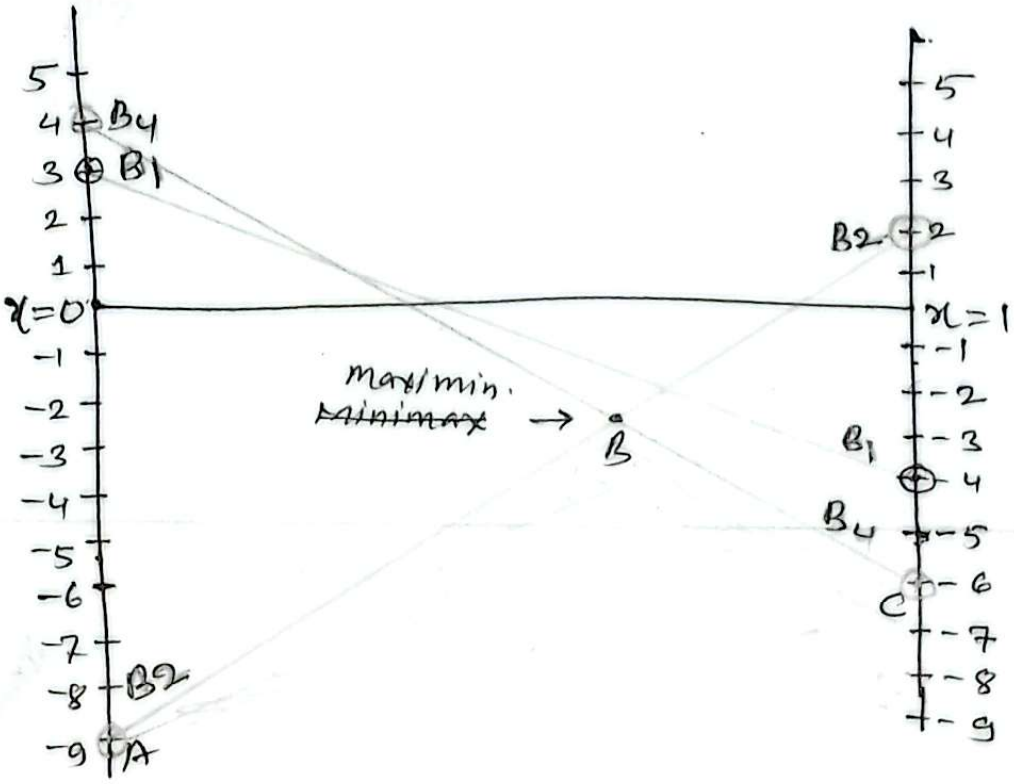
Higest value of column sum is 14, then column 5 is ≥ corresponding element of column 1, column 5 is dominated by column 1., delete column 5 similarly, column 1 is dominated by column 3, and delete column 3. ~~Also~~

~~Also, column 1 is dominated by column 4, & delete column 1~~

After column reduction, ~~2×2~~ pay-off matrix is

	Player B			Probability
	1	2	4	
Player A 1	-4	2	-6	x
2	3	-9	4	$1-x$

B's alternative	A's expected pay-off function	A's expected gain	
		$x=0$	$x=1$
1	$-4x+3(1-x) = -4x+3-3x = -7x+3$	3	−4
2	$2x+(-9)(1-x) = 2x-9+9x = 11x-9$	−9	2
4	$-6x+4(1-x) = -6x+4-4x = -10x+4$	4	−6



lower boundary points A, B and C. But point B is called maximin point.

Choose any two opposite ~~stop~~ slopes intersects at maximin point is B_2 and B_4.

	Player B 2	Player B 4	odd mens	Probability
Player A 1	2	−6	13	$P_1 = \frac{13}{13+8} = \frac{13}{21}$
Player A 2	−9	4	8	$P_2 = \frac{8}{21}$
odd mens	10	11		
Probability	$q_1 = \frac{10}{10+11} = \frac{10}{21}$	$q_2 = \frac{11}{21}$		

$\therefore$ value of game is $\frac{2\times10+(-6)\times11}{10+11} = \frac{20-66}{21} = \frac{-46}{21}$ R